Multiscale Random-Shape Convolution and Adaptive Graph Convolution Fusion Network for Hyperspectral Image Classification

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***Abstract*— Convolution neural networks (CNNs) are extensively utilized in hyperspectral image (HSI) classification due to their remarkable capability to extract features from patterns with fixed shapes. These networks have been shown to effectively capture features at the pixel level. However, the fixed shape of convolution kernels poses a challenge for CNNs to adapt to the diverse shapes found in HSIs. Graph neural networks (GNNs), particularly graph convolution networks (GCNs), possess robust feature extraction capabilities on graph structures and are extensively applied in HSI classification. However, one significant challenge in using GNNs is the selection of appropriate neighboring nodes for information aggregation. To address the existing challenges of GCN and CNN and leverage their respective advantages, this article introduces a novel patch-based CNN-GCN fusion classification network, named multiscale random-shape convolution and adaptive graph convolution fusion network (MRCAGCFN). It consists of a spectral transformation module (STM) and three main modules we proposed: a multiscale random-shape convolution (RSC) module for extracting convolution features, where the shape of the convolution kernel is randomized and a multiscale approach is applied to enhance adaptability to data with diverse shapes; an adaptive feature-fusion graph convolution module (AFGCM) for extracting graph convolution features, where the weights for neighborhood aggregation are learned adaptively to reduce feature fusion from dissimilar nodes and strengthen feature fusion from similar nodes; and an adaptive local feature processing module for processing features, where two different methods are employed to convert patch-level features to pixel-level features, thereby improving feature representation. MRCAGCFN combines the strengths of CNN and GCN while introducing enhancements to better accommodate diverse feature shapes. Experimental results on three HSI classification datasets demonstrate that our proposed MRCAGCFN outperforms some existing methods. The codes of our MRCAGCFN will be available at https://github.com/shengrunhua/MRCAGCFN.**

***Index Terms*— Convolution neural network (CNN), feature fusion, graph convolution network (GCN), hyperspectral image (HSI) classification.**

I. Introduction

With the advancement of remote sensing technology, remote sensing has found widespread applications in various fields such as agriculture and meteorology [1]. Hyperspectral images (HSIs) typically contain spectral information across hundreds of contiguous bands [2], and the rich spectral data, combined with the spatial information inherent in the images, proves valuable for tasks such as classification [3], target detection [4], change detection [5], and anomaly detection [6], [7]. Nowadays, the application of HSIs is a focal point of research in the field of remote sensing [8], [9].

The HSI classification involves categorizing every pixel in the entire image, which is akin to the segmentation task in regular images [10]. Due to the unique characteristics of HSIs, their rich spectral information allows for excellent differentiation of various materials [11]. However, the complexity of the spectral data simultaneously poses challenges in feature extraction [12], [13]. Due to the insufficient spatial resolution, the problems of same-object-different-spectrum and different-object-same-spectrum appear [14]. Initially, traditional classification methods were attempted in the field of HSI classification. Gualtieri and Chettri [15] proposed the use of a support vector machine (SVM) for HSI classification. Several handcrafted traditional classifiers were also applied to HSI classification, including k-nearest neighbors [16], random forest (RF) [17], sparse representation [18], and others. Researchers also attempted to enhance traditional methods. Fauvel et al. [19] proposed a method that involved processing and fusing both spatial and spectral information of the pixels to feed into an SVM classifier, Xia et al. [20] improved the RF method by introducing a classification approach that combined rotation forest (RoRF) with kernel principal component analysis (KCPA), known as RoRF-KPCA. At the same time, some traditional handcrafted methods such as linear discriminant analysis [21] and principal component analysis [22] have yielded good results in dimensionality reduction for HSIs. However, due to the limited feature extraction capabilities

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of traditional methods, they struggled to achieve satisfactory results in HSI classification. To enhance the performance of HSI classification, researchers began to explore methods with more robust feature extraction capabilities [23].

Due to the widespread application of deep learning (DL) in related fields such as natural language processing (NLP) [24] and computer vision [25]. In recent years, an increasing number of research teams have started to explore the use of DL networks in HSI classification tasks, achieving impressive results [26].

Convolution neural network (CNN) is one of the most common neural network models, using convolution kernels to extract features [27]. Due to the characteristic of weight sharing in convolution kernels, CNN has a strong capability to extract features with fixed shapes. In addition, due to the fixed nature of convolutional kernel shapes, conventional convolutions exhibit poor adaptability to irregularly shaped features. To enhance convolutional adaptability to diverse feature shapes, deformable convolutions have been introduced [28]. They achieve this by training the convolutional kernel shapes to align as closely as possible with the feature shapes. However, deformable convolutions determine their kernel shapes through training on the dataset, resulting in convolutional kernel shapes that are influenced by the feature shapes present in the training set. In hyperspectral data, where feature shapes vary widely, the spatial distribution of features in the training set may differ from that in the test set. This leads to challenges as deformable convolutions struggle to adapt their learned kernel shapes to the diverse feature shapes present in the test set, and the increased number of trainable parameters may elevate the risk of overfitting.

In addition to CNNs, Mou et al. [29] applied a recurrent neural network (RNN) with the gated recurrent unit (GRU) for HSI classification. They treated HSIs as sequential data for feature extraction and classification. Later, some researchers used long–short-term memory (LSTM) [30] for HSI classification. Subsequently, bidirectional LSTM (Bi-LSTM) was proposed, which constructed ordered sequences based on pixels, separately explored spatial and spectral features, and then concatenated them for classification [31]. To make sure the spatial feature and spectral feature are well explored by LSTM, Mei et al. [32] introduced an attention-based Bi-LSTM, HSI data were processed through an attention network before being fed into Bi-LSTM for feature extracting, simultaneously utilizing spatial and spectral information of a target pixel, and leading to improved classification performance. However, RNNs were originally designed to address tasks involving ordered sequences like NLP [33]. Adapting HSI information into suitable sequential data remains a significant challenge, and the mechanism for feature extraction based on sequences is not well-suited to accommodate features with various shapes.

Graph neural networks (GNNs) can perform feature extraction on graph data structures [34], which are suitable for representing features with various shapes. In HSI classification tasks, GNNs can effectively leverage the relationships between nodes formed by pixels or superpixels, making them highly capable of extracting features with various shapes. Graph convolution network (GCN), which is one of the most popular GNNs [35], is widely used for HSI classification. The GCN network initially used for HSI classification treats each pixel as a separate graph node [36], resulting in high computational and storage costs. To address this issue, methods like superpixel nodes [37] and mini-batch [38] have been applied in the hyperspectral domain. However, superpixel segmentation operations can lead to the loss of pixel-level features, and the mini-batch method only uses a randomly selected subset of nodes to construct the graph structure with the central node. It takes a fixed number of nodes as neighboring nodes for graph convolution, making it more prone to introducing features from dissimilar nodes and overlooking features from similar nodes. Especially for classes with fewer samples, the random selection of nodes in constructing the graph structure can easily include a significant number of dissimilar pixels, resulting in the loss of original features. To address the aforementioned issues, patch-based graph convolution has been applied to HSI classification [39]. This approach reduces computational costs while avoiding the loss of pixel-level features. All pixels within a patch participate in constructing the graph structure, enabling the network to adapt to the diverse shapes of features within the patch. However, existing patch-based GCNs have not been integrated with CNNs, thus failing to fully exploit the advantages of both. Due to the noise in the spectral dimension, existing patch-based GCNs relying solely on spectral features to construct the graph structure may not adequately reflect the correlation between nodes. Moreover, these methods struggle to leverage the characteristics of CNNs, which can progressively reduce feature sizes layer by layer, to effectively transform patch-level features into pixel-level features. This poses a challenging problem of how to reasonably perform this transformation. The strong adaptability of graph convolution to features with different shapes lies in its ability to perform neighborhood aggregation operations on graph structures. The key to improving the feature extraction capability of GCNs is how to select suitable neighboring nodes for neighborhood aggregation. Existing GCNs heavily rely on Laplacian matrices calculated based on feature differences to determine the aggregation weights for each node during the neighborhood aggregation operation. However, these GCNs often choose a fixed number of nodes as neighbor nodes for aggregation, inevitably introducing features from dissimilar nodes or neglecting features from similar nodes. This issue is particularly evident in patch-based GCNs, where the number of similar and dissimilar nodes varies significantly across different spatial positions in a patch. That makes selecting the same number of nodes as neighbors for aggregation poses challenges.

In summary, the challenges faced by existing HSI classification methods are as follows.

1) Relying solely on spectral features to construct the graph structure may not adequately capture the correlation between nodes due to noise in the spectral dimension.
2) The fixed shape of the convolution kernel results in poor adaptability to data with diverse feature shapes.
3) The neighborhood aggregation operation in GCNs easily introduces features from dissimilar nodes or overlooks

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features from similar nodes, impacting the spatial feature extraction capability.

4) Current patch-based GCNs use global pooling directly to convert patch-level features into pixel-level features. However, this conflicts with the logic of neighborhood aggregation in GCNs, leading to the introduction of erroneous information.

To address the above problems, this article introduces a new method called multiscale random-shape convolution and adaptive graph convolution fusion network (MRCACGFN) for HSI classification. MRCACGFN comprises a spectral transformation module (STM) and three main modules we propose, a CNN-based feature extraction module named multiscale random-shape convolution module (MRCM), a GCN-based feature extraction module called adaptive feature-fusion graph convolution module (AFGCM), and a feature processing module called adaptive local feature preprocessing module (ALFPM). The random-shape convolution (RSC) and adaptive feature-fusion graph convolution (AFGC) are proposed in our work to enhance the feature extraction capabilities of CNNs and GCNs. The feature processing module we introduce effectively transforms patch features into the correct central pixel features.

Our major contributions are as follows.

1) We introduce a novel neural network for HSI classification, fusing patch-based GCN and CNN. This integration mitigates the issue of the graph structure relying on spectral features and not effectively reflecting the correlation between nodes.
2) We enhance the adaptability of the convolution layer to data with diverse feature shapes by introducing randomness to the convolution kernel shapes.
3) We improve the neighborhood aggregation operation of GCNs by adaptively learning different weights for different nodes. This reduces the introduction of information from dissimilar nodes while preventing the discarding of too much information from similar nodes. In addition, it utilizes spatial position information to alleviate graph structural errors caused by spectral noise.
4) Leveraging spatial aggregation concepts from both CNN and GCN, we efficiently transform patch-level features into pixel-level features, enhancing feature representation in the classification process.

The remainder of this article is organized as follows. Section II introduces some research works relevant to the direction of our study. Section III provides detailed implementation details of the proposed MRCAGCFN. Section IV covers the experimental part, presenting the details and results of a series of validation experiments based on this method. Finally, Section V concludes the article.

II. Related Work

In this section, we review some representative studies in the field of HSI classification that are relevant to this article.

A. Convolution Neural Networks

Due to the property of weight sharing in convolutional kernels, CNNs exhibit strong feature learning capabilities and are widely applied in HSI classification tasks. Roy et al. [40] introduced a HybirdSN, which concatenates 2-D and 3-D convolutions for HSI classification, extracting both spatial and spectral features by leveraging the advantages of both 2-D and 3-D convolutions. Zhong et al. [41] proposed a spectral–spatial residual network (SSRN), which not only utilizes 3-D convolutions for spatial and spectral feature extraction but also employs residual networks to alleviate the gradient vanishing problem associated with deep networks, successfully increasing the network's depth. Since different spectral dimensions and spatial positions have varying impacts on HSI classification, attention mechanisms have also been widely applied. Roy et al. [42] introduced an attention-based adaptive spectral–spatial kernel ResNet (A2S2K—ResNet), which builds upon SSRN by incorporating attention mechanisms for spectral and spatial information, thereby improving classification accuracy. Li et al. [43] presented a double-branch dual-attention mechanism network (DBDA), which uses two branches with attention mechanisms for spectral and spatial information, enhancing feature learning. The methods mentioned above extract spatial–spectral features simultaneously, hierarchically, or separately. Chen et al. [44] proposed a grid network (GNet), which incorporates the concept of spectral–spatial cross-layer fusion. This approach fuses the spectral and spatial features extracted by different depths and types of convolution layers, addressing the problem of unbalanced feature representation, and improving classification accuracy. However, CNN's fixed-shape convolution kernels make it challenging to extract features with different shapes effectively. To address the problem of extracting features with different shapes, Zhu et al. [45] proposed a deformable HSI classification network (DHCNet). It uses deformable convolutions to adaptively change the shapes of convolution kernels for extracting features with different shapes. However, the shapes of deformable convolution kernels remain fixed after training, and these shapes may not necessarily adapt well to the features in the testing set. Bai et al. [46] proposed an adaptive dual attention network (ADAN), which can extract adjacent superpixel features while extracting adjacent pixel features, enhancing the model's adaptability to irregular shapes.

B. Graph Convolution Networks

GCNs can perform feature extraction on graph data structures, demonstrating strong adaptability to data with diverse feature shapes. Mou et al. [36] proposed a nonlocal GCN, which uses all pixels of HSI to construct the graph data structure. Residential connection is also used in GCNs to form deeper GCN [47]. Bai et al. [47] improved a graph structure and enhanced network performance by employing attention mechanisms. Duan et al. [48] introduced a structure-preserved hyper GCN (SPHGCN), which built a hypergraph structure to extract features instead of a normal graph structure. However, GNNs have limitations. During the graph convolution process, all data nodes need to be involved in constructing the graph, and training requires a global view of the data. When dealing with a large number of nodes, the computational and storage costs can become prohibitively high. In addition, the fixed

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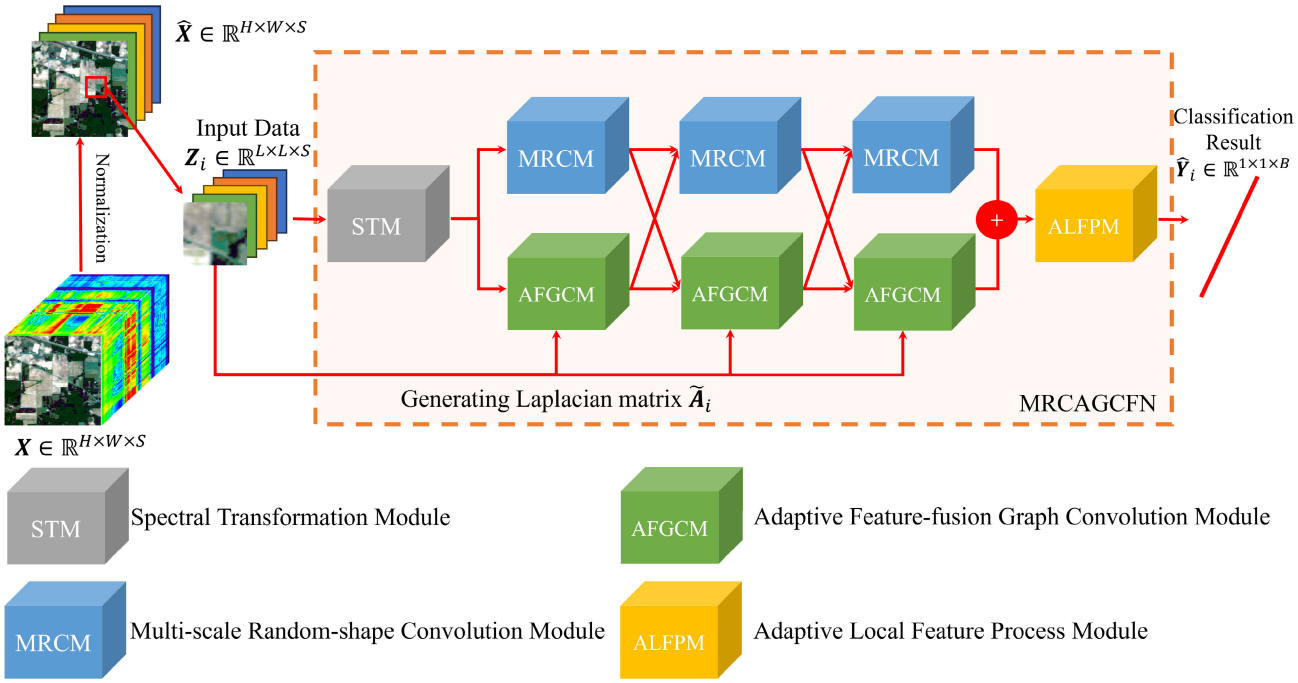


Fig. 1. Overview of our proposed MRCAGCFN.

graph scale means that retraining the network is necessary when making predictions for new samples. Wan et al. [37] proposed a multiscale dynamic GCN, which used the simple linear iterative cluster (SLIC) superpixel segmentation method [49] to reduce the number of graph nodes and reduce computational and storage costs, the article also introduced the method of multiscale dynamic graph. Building upon the previous content, Wan et al. [50] optimized the features of superpixel regions. They utilized the similarity among pixels within a region, assigning different feature weights to pixels within the region to make the superpixel features more reasonable. Bai et al. [51] proposed a graph attention network with an adaptive graph structure mining approach (GAT-AGSM), which improves the graph structure and superpixel representation by utilizing the adaptive graph structure mining mechanism and the attention mechanism. Liu et al. [52] proposed a CNN-enhanced GCN (CEGCN), which combines CNN and GCN for classification. This approach leverages the strengths of CNN for extracting features with regular shapes and GCN for handling complex and diverse feature shapes. Zhou et al. [53] introduced an attention multihop graph and multiscale convolution fusion network (AMGCFN), which used an attention mechanism to fuse pixel-level features from the CNN branch and superpixel-level features from the GCN branch to improve classification accuracy. Chen et al. [54] proposed a local aggregation and global attention network (LAGAN), which uses spectral-induced aligned superpixel segmentation and graph transformer to improve the representation of the graph. Except for GCNs, edge convolution is also used with SLIC for HSI classification [55]. However, these networks are still trained globally, they fail to predict the new input samples without retraining the network. To address this issue, Hong et al. [38] proposed a minibatch GCN (miniGCN), which performs training on subgraphs. This allows training with subgraph-based networks, it avoids the need for retraining when predicting new samples. Zhang et al. [39] introduced a spatial pooling GCN (SPGCN), they utilize a patch-based approach to construct pixel-level graph structures within fixed-sized patches.

III. Proposed Method

In this section, we provide a detailed introduction to our proposed classification network, MRCAGCFN.

A. Overview of Proposed MRCAGCFN

The architecture of the network is shown in Fig. 1. In Fig. 1, HSI data is represented as a data cube, denoted as $\mathbf{X} \in \mathbb{R}^{H\times W\times S}$, where H and W are its spatial height and width, and S is the number of spectral channels. In the preprocessing stage, we start by performing global normalization on the data, the data after normalization is represented as $\hat{\mathbf{X}} \in \mathbb{R}^{H\times W\times S}$ in Fig. 1. After that, we proceed to divide the normalized data $\hat{\mathbf{X}}$ into patches, which are centered around the pixels to be classified and used as input data for the proposed network, and the processed dataset is represented as $\mathcal{Z} = \{\mathbf{Z}_1, \mathbf{Z}_2, \ldots, \mathbf{Z}_n\}, \mathbf{Z}_i \in \mathbb{R}^{L\times L\times S}$ and $\mathcal{Y} = \{\mathbf{Y}_1, \mathbf{Y}_2, \ldots, \mathbf{Y}_n\}, \mathbf{Y}_i \in \mathbb{R}^{1\times 1\times B}$, where $\mathcal{Z}$ represents the input data,, $\mathcal{Y}$ represents the one-hot labels for the samples, n is the total number of samples for pixels that need to be classified, L is the side length of the patches, and B is the number of sample classes. $\mathbf{Z}_i$ in Fig. 1 represents one element from the input data $\mathcal{Z}$.

Finally, we construct a pixel-level graph structure using the data partitioned into patches. In our patch-based graph structure, every pixel in the patch serves as a graph node, and the

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graph structure can be represented as $\mathcal{G} = \{G_1, G_2, \ldots, G_i = \{\mathbf{Z}_{\text{if}}, \tilde{\mathbf{A}}_i\}, \ldots, G_n\}$, where G_i is the pixel-level graph structure built from $\mathbf{Z}_i$. $\mathbf{Z}_{\text{if}} \in \mathbb{R}^{L^2 \times S}$ is the set of vertices, which is the 1-D arrangement of spatial pixels from $\mathbf{Z}_i$. $\tilde{\mathbf{A}}_i \in \mathbb{R}^{L^2 \times L^2}$ represents the set of edges in the graph, and it is computed based on $\mathbf{Z}_i$, as shown in Fig. 1. $\tilde{\mathbf{A}}_i$ is derived from the graph's adjacency matrix $\mathbf{A}_i \in \mathbb{R}^{L^2 \times L^2}$, which is based on the spectral feature differences between nodes (pixels). The formula for constructing $\mathbf{A}_i$ is given as

$$\mathbf{A}_i(a, b) = \exp\left(-\frac{\|\mathbf{Z}_{\text{if}}(a) - \mathbf{Z}_{\text{if}}(b)\|^2}{S}\right) \tag{1}$$

where $a, b = 1, 2, \ldots, L^2$ represent the spatial coordinates of points in $\mathbf{Z}_{\text{if}}$.

After obtaining the adjacency matrix, we calculate the Laplacian matrix $\tilde{\mathbf{A}}_i$ as the set of edges for the graph. $\tilde{\mathbf{A}}_i$ is calculated by the formula is defined as

$$\tilde{\mathbf{A}}_i = \mathbf{I} + \mathbf{D}_i{}^{-1}\mathbf{A}_i \tag{2}$$

where $\mathbf{D}_i \in \mathbb{R}^{L^2 \times L^2}$ is the degree matrix of $\mathbf{A}_i$, calculated as $\mathbf{D}_i(a, a) = \sum_b \mathbf{A}_i(a, b)$, and $\mathbf{I} \in \mathbb{R}^{L^2 \times L^2}$ is the identity matrix.

After all the preprocessing steps are completed, we feed the training data and their corresponding graph structures into our proposed network for parameter training. The network we propose primarily employs one STM, three MRCMs, three AFGCMs, and one ALFPM, as shown in Fig. 1.

Before feature extraction, we first apply dimensionality reduction to the hyperspectral data by using STM, which mainly consists of two 1×1 convolutions. As Fig. 1 shows, in the feature extraction phase, the input data for each layer is fed into both MRCM and AFGCM for individual feature extraction, and the features extracted from both modules are averaged before they are passed to the next layer of the network. The fusion approach ensures that the input data for each layer benefits from the feature extraction capabilities of both modules, preventing the imbalance in HSI features from affecting the network's performance.

After feature extraction is completed, the extracted features are sent to ALFPM for feature processing, as Fig. 1, which transforms these features into per-class prediction probabilities for the central pixels $\hat{\mathcal{Y}} = \{\hat{\mathbf{Y}}_1, \hat{\mathbf{Y}}_2, \ldots, \hat{\mathbf{Y}}_n\}$, $\hat{\mathbf{Y}}_i \in \mathbb{R}^{1 \times 1 \times B}$.

We calculate the cross-entropy loss between the output classification results and the one-hot encoded labels, then reduce the loss through gradient descent to learn the model parameters. The formula for the cross-entropy loss is

$$L = \frac{1}{N}\sum_{i=0}^{N-1} L_i = -\frac{1}{N}\sum_{i=0}^{N-1}\sum_{c=0}^{B-1} \mathbf{Y}_i(c)\log \hat{\mathbf{Y}}_i(c) \tag{3}$$

where L is the total loss, L_i is the loss for the ith sample, and N is the number of samples. $\mathbf{Y}_i$ is the one-hot label of the ith sample, and $\hat{\mathbf{Y}}_i(c)$ is the predicted probability of the ith sample belonging to class c.

In the upcoming part of this section, we will provide a detailed explanation of the STM and the three main modules we propose.

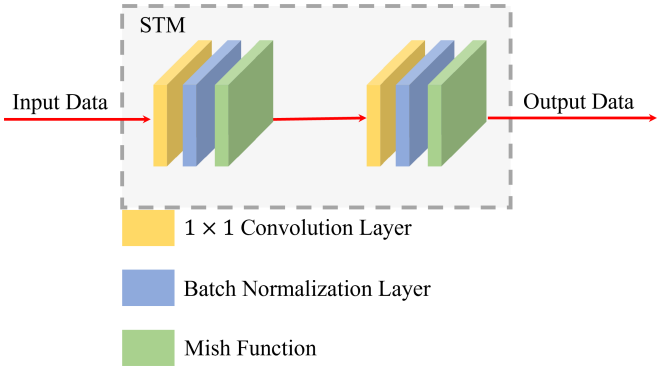


Fig. 2. Architecture of STM.

B. Spectral Transformation Module

The high dimensionality of HSI introduces a lot of redundant information. To reduce redundant information in the spectra, we employ an STM which mainly consists of two 1×1 convolutions. The details of STM are shown in Fig. 2, and it can be defined by the formula as

$$\mathbf{Z}_i^{\text{spe}} = \text{Mish}\big(\text{BN}\big(\text{Mish}\big(\text{BN}\big(\mathbf{Z}_i\mathbf{W}^{\text{sp1}} + \mathbf{b}^{\text{sp1}}\big)\big)\mathbf{W}^{\text{sp2}} + \mathbf{b}^{\text{sp2}}\big)\big) \tag{4}$$

where $\mathbf{Z}_i \in \mathbb{R}^{L \times L \times S}$ represents the input data of MRCAGCFN and $\mathbf{Z}_i^{\text{spe}}$ is the output of STM. $\mathbf{W}^{\text{sp1}} \in \mathbb{R}^{S \times m}$ and $\mathbf{W}^{\text{sp2}} \in \mathbb{R}^{m \times n}$ are the parameters matrixes of STM and $\mathbf{b}^{\text{sp1}}$ and $\mathbf{b}^{\text{sp2}}$ are the biases.

BN($\bullet$) represents the batch normalization layer, and the formula is defined as

$$\hat{\mathbf{T}} = \frac{\mathbf{T} - \text{E}(\mathbf{T})}{\sqrt{\text{Var}(\mathbf{T}) + \epsilon}} \times \gamma + \beta \tag{5}$$

where $\mathbf{T} \in \mathbb{R}^{L \times L \times n}$ is the input data of the batch normalization layer and $\hat{\mathbf{T}} \in \mathbb{R}^{L \times L \times n}$ is the output after applying batch normalization. E($\bullet$) represents the mean function, Var($\bullet$) is the variance function, and ϵ is a small correction term to prevent the denominator from becoming too small. γ and β are trainable parameters used to adjust the mean and variance, ensuring that the data after batch normalization closely matches the original distribution.

Mish($\bullet$) represents the mish activation function, the formula for mish as

$$\text{Mish}(x) = x \times \tanh\big(\ln\big(1 + e^x\big)\big) \tag{6}$$

where x is the input data and Mish(x) represents the output data.

C. Multiscale RSC Module

While CNNs have been extensively utilized in HSI classification and have demonstrated strong feature extraction capabilities, they face limitations due to the fixed square shape of conventional convolutions. In HSIs, land cover is highly various and exhibits significant shape variations, making it challenging for standard convolutions to adapt to the various

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land cover scenarios in HSI classification data. The formula for a standard convolution operation can be defined as

$$\mathbf{y}(p_0) = \sum_{p_n \in R} \mathbf{w}(p_n) \cdot \mathbf{x}(p_0 + p_n) \tag{7}$$

where p_0 represents a pixel in the feature map and p_n is one of the spatial offset values in the convolution kernel's spatial domain R. $p_0 + p_n$ gives one of the sampling pixels within the convolution kernel's size range around p_0, $\mathbf{y}(p_0)$ is the new value of p_0 after feature extraction, $\mathbf{x}(p_0 + p_n)$ is the old value of $p_0 + p_n$ before feature extraction, and $\mathbf{w}(p_n)$ is the weight.

To enable CNNs to extract features of various shapes, deformable convolutions have been introduced and applied in the field of DL. Deformable convolutions introduce an additional parameter called the offset on top of standard convolutions, allowing the shape of the convolution kernel to change, and enabling the convolution kernel to extract features of different shapes. However, the shapes of the convolution kernels obtained through training in deformable convolutions remain fixed, and the learned convolution kernel shapes from the training set may not adapt well to the feature shapes of the testing set.

To enhance the adaptability of CNNs to features of various shapes, we propose a novel convolution technique called RSC. Specifically, we employ a method of randomly generating offsets to make the convolution kernel's shape random during the convolution operation. RSC randomly selects sampling positions and extracts their pixel features for the convolution operation, and we use bilinear interpolation to address the issue of convolution sampling positions having no corresponding pixels after applying offsets. This operation ensures that the shape of each convolution kernel is not fixed, addressing the issue of fixed convolution kernel shape and allowing the kernels to better adapt to various feature shapes, enhancing the network's robustness. To prevent excessive randomness that could lead to feature loss, we impose restrictions on the random generation of offset values, ensuring that the randomly sampled convolution kernel areas remain close to the original convolution region.

The RSC adds an offset Δp_n to standard convolution, where Δp_n is randomly generated within our specified constraints, making the convolution kernel shape random. The formula of RSC as

$$\mathbf{y}(p_0) = \sum_{p_n \in R} \mathbf{w}(p_n) \cdot \mathbf{x}(p_0 + p_n + \Delta p_n) \tag{8}$$

where Δp_n is the random for different pixels in the feature map. The difference between standard convolution and RSC is shown in Fig. 3.

Based on RSC, we designed a convolution feature extract module named MRCM. The architecture of our proposed MRCM is shown in Fig. 4. In this module, we utilize RSC for feature extraction and incorporate three different scale RSC layers (3×3, 5×5, and 7×7) in parallel to better extract features of different scales. Each of these three different scale RSC layers individually selects random sampling positions within different scale ranges and convolves them to extract features, that further improve the model's robustness. After

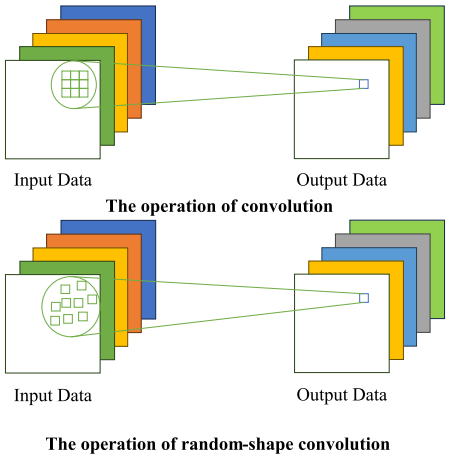


Fig. 3. Operation of standard convolution and RSC.

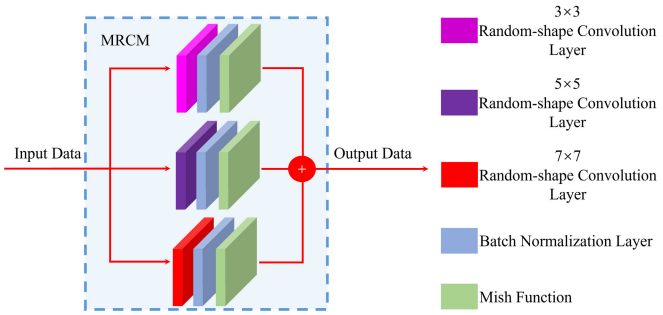


Fig. 4. Architecture of our proposed MRCM.

each RSC layer, there is a batch normalization layer to process the data and accelerate model training. Following the batch normalization layer, we use the Mish activation function to introduce nonlinearity to the data. Finally, the features extracted from the three branches are averaged to obtain the final output feature. MRCM we propose can be expressed by the formula as

$$\begin{aligned}\mathbf{Z}_i^{(l+1)}(x, y, k) &= \frac{1}{3}\sum_{i=1}^{3} \text{Mish}\left(\text{BN}\left(\sum_{j=0}^{n-1}\sum_{h=0}^{S_i-1}\sum_{w=0}^{S_i-1} \mathbf{W}_{i,k,j}^{(l)}(h, w)\right.\right. \\ &\quad \left.\left. \cdot \mathbf{Z}_i^{(l)}(x + h + \Delta h_i, y + w + \Delta w_i, j) + \mathbf{b}_{i,k}^{(l)}\right)\right)\end{aligned} \tag{9}$$

where $\mathbf{Z}_i^{(l)} \in \mathbb{R}^{L \times L \times n}$ represents the input data cube for the MRCM of the lth layer, and $\mathbf{Z}_i^{(l+1)} \in \mathbb{R}^{L \times L \times n}$ is the output data for the MRCM of the lth layer. $\mathbf{Z}_i^{(l+1)}(x, y, k)$ is the value of the point at coordinates (x, y, k) in the output data cube, k represents a channel dimension, and the features in this dimension are extracted by the kth convolution kernel from all three convolution layers. $\mathbf{W}_{i,k,j}^{(l)} \in \mathbb{R}^{S_i \times S_i}$ is the parameter matrix corresponding to the jth channel of the kth convolution kernel in the ith convolution layer of the MRCM of the lth layer, S_i is the side length of the convolution kernel in the ith convolution layer, and $\mathbf{b}_{i,k}^{(l)}$ is the bias.

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$\mathbf{Z}_i^{(l)}(x+h+\Delta h_i, y+w+\Delta w_i, j)$ is one of the sampling positions in the input data cube for the ith RSC layer, and Δh_i and Δw_i are random offset values.

D. AFGC Module

The graph is a non-Euclidean data structure composed of nodes and edges. GCNs allow for feature extraction on data structured as graphs, a capability that traditional CNNs cannot achieve. Specifically, GCNs aggregate information from a node and its neighboring nodes, enabling it to extract node features based on the relationships between nodes in the graph structure, which makes GCNs well-suited for extracting features of various shapes.

GCNs also introduce certain challenges. When the graph structure is too large, an excessive number of graph nodes can lead to increased computational costs. To address this issue, we adopt a method of constructing pixel-level graph structures on patches for graph convolution operations. This approach not only reduces the number of nodes in a single training instance but also aids in predicting new samples without the need for repeated training. Compared to super-pixel segmentation methods, the patch-based approach avoids the loss of pixel-level features.

The ability to aggregate information from neighboring nodes is the fundamental reason for the adaptability of GCNs to features of various shapes and effectively aggregating information from neighboring nodes is crucial for improving the effectiveness of GCNs. Existing GCNs often aggregate information from a fixed number of nodes as neighbors, which can lead to the inclusion of heterogeneous node information or the exclusion of homogeneous node information and makes it challenging to select an appropriate number of neighboring nodes. To address this issue, we propose an improved graph convolution layer named AFGC.

In the AFGC, the input data undergo feature extraction first, similar to the traditional graph convolution layer. Then, we design a method called adaptive neighborhood aggregation. Specifically, we first perform neighborhood aggregation based on the Laplacian matrix. After that, the data that has undergone neighborhood aggregation is adaptively fused with the data that has not undergone neighborhood aggregation in terms of channel dimensions, we refer to this approach as adaptive feature-fusion. Due to the differences in the neighborhood situations for different nodes, the information fusion weights for different nodes are also different, which allows for better adaptation to the neighborhood conditions of different nodes, resulting in more reasonable learned neighborhood aggregation weights. Based on AFGC, we construct the graph convolution feature extract module called AFGCM. The architecture of our proposed AFGCM and details of the AFGC are shown in Fig. 5, AFGCM can also be represented by the formula as

$$\mathbf{Z}_{\text{if}}^{(l+1)} = \text{Mish}\Big(\text{BN}\Big(\tilde{\mathbf{A}}_i\Big(\mathbf{Z}_{\text{if}}^{(l)}\mathbf{W}_{\text{gcn}}^{(l)} + \mathbf{b}^{(l)}\Big) \otimes \mathbf{W}_{f1}^{(l)} + \Big(\mathbf{Z}_{\text{if}}^{(l)}\mathbf{W}_{\text{gcn}}^{(l)} + \mathbf{b}^{(l)}\Big) \otimes \mathbf{W}_{f2}^{(l)}\Big)\Big) \quad (10)$$

where $\mathbf{Z}_{\text{if}}^{(l)} \in \mathbb{R}^{L^2 \times n}$ represents the input data for the AFGCM of the lth layer, $\mathbf{Z}_{\text{if}}^{(l+1)} \in \mathbb{R}^{L^2 \times n}$ is the output data for the AFGCM of the lth layer. $\mathbf{W}_{\text{gcn}}^{(l)} \in \mathbb{R}^{n \times n}$ is the graph convolution parameter matrix for the AFGCM of the lth layer, and $\mathbf{b}^{(l)}$ is the bias. $\tilde{\mathbf{A}}_i$ the Laplacian matrix corresponding to the input data. $\mathbf{W}_{f1}^{(l)} \in \mathbb{R}^{L^2 \times 1}$ and $\mathbf{W}_{f2}^{(l)} \in \mathbb{R}^{L^2 \times 1}$ are two parameter matrices for adaptive feature-fusion in the AFGCM of the lth layer. $\otimes$ denotes element-wise multiplication.

E. Adaptive Local Feature Processing Module

HSI classification tasks are performed at the pixel level, and the features extracted within patches need to be transformed into features for central pixels, as they are used for final classification. For super-pixel GCN methods, the typical approach is to directly use super-pixel features corresponding to the central pixel for the final prediction. For patch-based methods, using pooling techniques to transform patch features into central pixel features is a common approach. However, this method can lead to the mixing of information from heterogeneous pixels in patches. To solve this problem, Some CNN-based methods leverage the property of convolution layers to gradually reduce input size, ultimately resulting in patches of only 1×1 in size, effectively converting patch features into central pixel features. However, pixel-level GCN methods cannot exploit the same property of convolution layers to reduce patch size. Some methods use attention mechanisms to process features after feature extraction to improve feature representation. Still, the distribution of pixel categories within different patches can vary significantly, making this approach less effective for handling the various feature shapes present in HSIs. SPGCN uses a layer-by-layer pixel exclusion process and average pooling to aggregate patch-shaped features into central pixel features. However, the operation of excluding outer pixels layer by layer may risk discarding features of homogeneous pixels, and average pooling could introduce features from heterogeneous pixels. To address this challenge, we propose a feature processing module called ALFPM. The architecture of ALFPM is shown in Fig. 6.

Our proposed ALFPM can be divided into two main components. One component is designed to process features with relatively regular shapes using depthwise convolution, which is named the depthwise convolution feature processing component. Another component involves adaptive neighborhood aggregation to adapt to the varied shapes of HSI features, which is named the adaptive neighborhood aggregation feature processing component.

In the depthwise convolution feature processing component, ALFPM employs depthwise convolution which is group convolution with the same number of input and output channels as well as the same number of groups. The kernel size of the depthwise convolution in this component is the same as the patch size to process data into 1×1 pixel features. The use of depthwise convolution is aimed at reducing the number of parameters while still enabling the learning of different parameters across various channels and helps save computational time and guards against overfitting due to an excessive number of parameters.

In the adaptive neighborhood aggregation feature processing component, ALFPM leverages the concept of adaptive neighborhood aggregation we introduced earlier. Without feature

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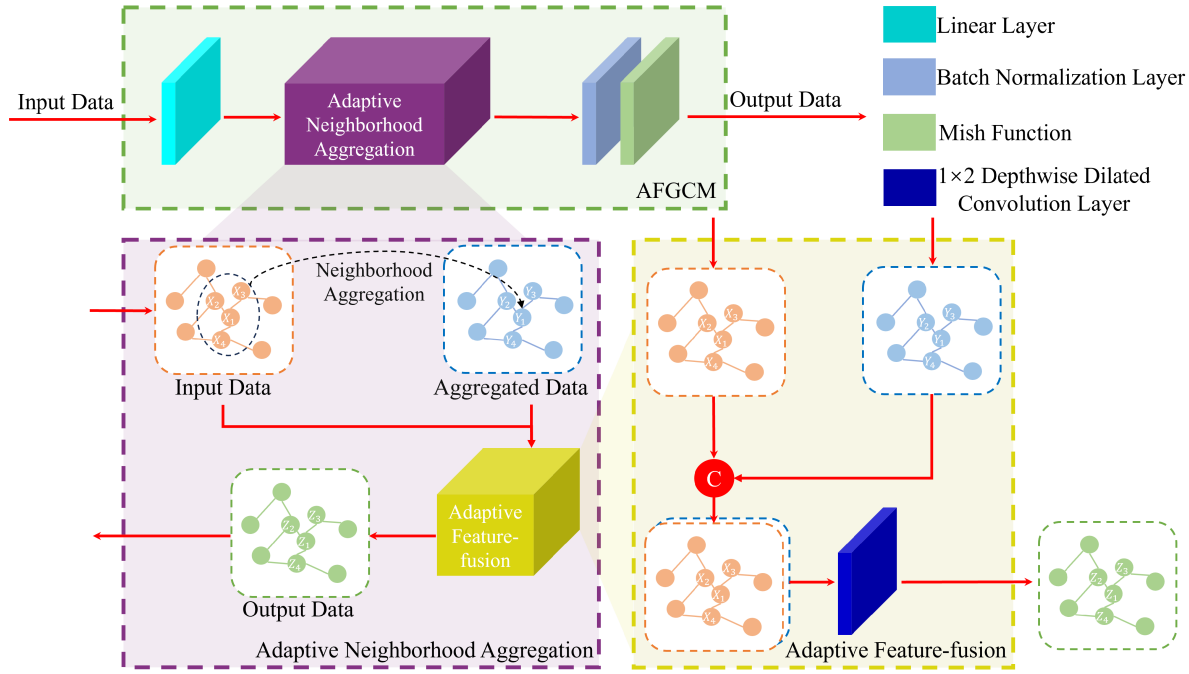


Fig. 5. Architecture of our proposed AFGCM and details of the adaptive neighborhood aggregation method.

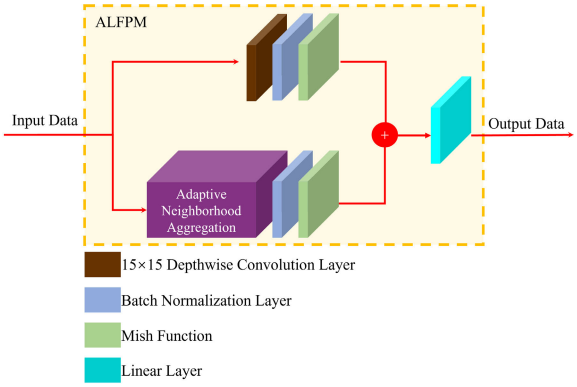


Fig. 6. Architecture of our proposed ALFPM.

extraction, ALFPM directly performs adaptive neighborhood aggregation and extracts central pixel information from the aggregated results. Neighborhood aggregation takes advantage of the graph structure, allowing it to adapt well to various feature shapes, adaptive feature-fusion further enhances its adaptability to different pixel distribution patterns.

The depthwise convolution feature processing component of our proposed ALFPM can be represented by a formula as

$$\hat{\mathbf{Y}}_i^{\mathrm{Cov}}(1,1,j) = \mathrm{Mish}\left(\mathrm{BN}\left(\sum_{h=0}^{L-1}\sum_{w=0}^{L-1}\mathbf{W}_{\mathrm{Cov},j}(h,w) \cdot \mathbf{Z}_i^{\mathrm{out}}(x+h, y+w, j) + \mathbf{b}_{\mathrm{Cov},j}\right)\right) \quad (11)$$

where $\hat{\mathbf{Y}}_i^{\mathrm{Cov}} \in \mathbb{R}^{1\times1\times n}$ represents the output data of the depthwise convolution feature processing component. $\mathbf{W}_{\mathrm{Cov},j} \in \mathbb{R}^{L\times L}$ is the parameter matrix of the jth convolution kernel which computes on the jth channel, and $\mathbf{b}_{\mathrm{Cov},j}$ is the bias. $\mathbf{Z}_i^{\mathrm{out}} \in \mathbb{R}^{L\times L\times n}$ is the input data of the depthwise convolution feature processing component.

The adaptive neighborhood aggregation feature processing component of our proposed ALFPM can be represented by a formula as

$$\hat{\mathbf{Y}}_i^{\mathrm{Adt}} = \tilde{\mathbf{A}}_i\left(:, \frac{L^2-1}{2}\right)\mathbf{Z}_{\mathrm{if}}^{\mathrm{out}} \otimes \mathbf{W}_{f1}^{\mathrm{out}} + \mathbf{Z}_{\mathrm{if}}^{\mathrm{out}}\left(\frac{L^2-1}{2}, :\right) \otimes \mathbf{W}_{f2}^{\mathrm{out}} \quad (12)$$

where $\hat{\mathbf{Y}}_i^{\mathrm{Adt}} \in \mathbb{R}^{1\times n}$ is the output data of the adaptive neighborhood aggregation feature processing component. $\mathbf{Z}_{\mathrm{if}}^{\mathrm{out}} \in \mathbb{R}^{L^2\times n}$ represents the input data of the adaptive neighborhood aggregation feature processing component, $\tilde{\mathbf{A}}_i(:, ((L^2-1)/2)) \in \mathbb{R}^{1\times L^2}$ is the Laplacian matrix corresponding to the central pixel of $\mathbf{Z}_{\mathrm{if}}^{\mathrm{out}}$, which is represented by $\mathbf{Z}_{\mathrm{if}}^{\mathrm{out}}(((L^2-1)/2), :) \in \mathbb{R}^{1\times n}$. $\mathbf{W}_{f1}^{\mathrm{out}} \in \mathbb{R}^{1\times1}$ and $\mathbf{W}_{f2}^{\mathrm{out}} \in \mathbb{R}^{1\times1}$ are the parameters for adaptive feature-fusion.

The two components process the patch features to obtain the central pixel features, which are then averaged to obtain the improved central pixel features. Finally, these features are passed through a fully connected layer for the final classification prediction. This operation can be represented by a formula as

$$\hat{\mathbf{Y}}_i = \sigma\left(\frac{1}{2}\left(\hat{\mathbf{Y}}_i^{\mathrm{Cov}} + \hat{\mathbf{Y}}_i^{\mathrm{Adt}}\right)\mathbf{W}_{Fc}^{\mathrm{out}} + \mathbf{b}_{Fc}^{\mathrm{out}}\right) \quad (13)$$

where $\hat{\mathbf{Y}}_i \in \mathbb{R}^{1\times1\times B}$ is the final output data of MRCAGCFN, which represents the probability predictions for the central pixel belonging to various classes. $\sigma(\bullet)$ is the Softmax activation function. $\mathbf{W}_{Fc}^{\mathrm{out}} \in \mathbb{R}^{n\times B}$ is the parameter matrix of the fully connected layer, and $\mathbf{b}_{Fc}^{\mathrm{out}}$ represents the bias.

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IV. Experiments

In this section, we provide detailed descriptions of a series of experiments conducted to validate the effectiveness of our proposed MRCAGCFN.

A. Dataset Description

In our experiments, we evaluated the performance of our proposed MRCAGCFN by using three widely used HSI datasets: Indian Pines (IP), Pavia University (PU), and Huston University 2013 (HU).

1) The IP dataset was acquired by the AVIRIS sensor over an agricultural test site in northwestern Indiana. It consists of 145 × 145 pixels with a spatial resolution of 20 m and 220 spectral dimensions. The area mainly contains 16 classes with a total of 10 366 labeled samples. The modified Indiana Pines dataset used in our experiment reduced the number of bands to 200 by removing interfering bands.
2) The PU dataset was collected over Pavia, Northern Italy, using the ROSIS sensor. It consists of 610 × 340 pixels with a spatial resolution of 1.3 m and 103 spectral dimensions. This dataset contains nine classes with a total of 43 923 labeled samples.
3) The HU dataset was collected in June 2012 by the CASI sensor over the University of Houston campus. It comprises 349 × 1905 pixels with a spatial resolution of 2.5 m and 144 spectral dimensions. This dataset includes 15 categories with a total of 15 029 labeled samples.

Some researchers argued that in HSI classification tasks, partitioning patches around pixels to be classified might result in the inclusion of other pixels to be classified in the patch, simplifying the classification task. Liang et al. [56] proposed using a partitioning method that enhances the separability of the training and testing sets to increase the classification difficulty. On the IP and PU datasets, we adopted their partitioning method, and our training and testing set distributions align with some methods such as GNet [44] and miniGCN [38]. Such an approach increases the difference in feature distribution between the training set and the test set, thereby adding difficulty to the classification task and introducing a greater risk of overfitting.

Moreover, in the realm of machine learning, it is desirable for the feature distribution in the training set to closely resemble that of the test set. Some researchers argue that employing a random selection of training samples aligns better with the expectations of machine learning. Many studies in the field of HSI classification opt for the approach of randomly selecting samples for training and testing sets, aiming to evaluate a model's feature learning capability under ideal conditions. In line with this, we randomly selected 5% of the total samples as training samples in the HU dataset. By utilizing various dataset partitioning methods, we aim to showcase the effectiveness of our approach across datasets with diverse characteristics. To maintain consistency with the official dataset splits, we did not create a validation set for the HU dataset.

TABLE I
Number of Training and Testing Samples on the IP

No.	Class	Training	Testing
1	Corn Notill	50	1384
2	Corn Mintill	50	784
3	Corn	50	184
4	Grass Pasture	50	447
5	Grass Trees	50	697
6	Hay Windrowed	50	439
7	Soybean Notill	50	918
8	Soybean Mintill	50	2418
9	Soybean Clean	50	564
10	Wheat	50	162
11	Woods	50	1244
12	Buildings Grass Trees Drives	50	330
13	Stone Steel Towers	50	45
14	Alfafa	15	39
15	Grass Pasture Mowed	15	11
16	Oats	15	5
	Total	695	9671

TABLE II
Number of Training and Testing Samples on the PU

No.	Class	Training	Testing
1	Asphalt	548	6304
2	Meadows	540	18146
3	Gravel	392	1815
4	Trees	524	2912
5	Metal Sheets	265	1113
6	Bare Soil	532	4572
7	Bitumen	375	981
8	Bricks	514	3364
9	Shadows	231	795
	Total	3921	40002

TABLE III
Number of Training and Testing Samples on the HU

No.	Class	Training	Testing
1	Healthy Grass	63	1188
2	Stressed Grass	63	1191
3	Synthetic Grass	35	662
4	Tree	63	1181
5	Soil	63	1179
6	Water	17	308
7	Residential	64	1204
8	Commercial	63	1181
9	Road	63	1189
10	Highway	62	1165
11	Railway	62	1173
12	Parking Lot1	62	1171
13	Parking Lot2	24	445
14	Tennis Court	22	406
15	Running Track	33	627
	Total	759	14270

Tables I–III provide the number of samples in the training and test sets after splitting for the three datasets, while Fig. 7 illustrates the spatial distribution of the training and test samples for each dataset.

B. Experimental Setup

Our proposed MRCAGCFN is conducted by using Python 3.8.18 and PyTorch 2.0.1 on an Ubuntu system. The

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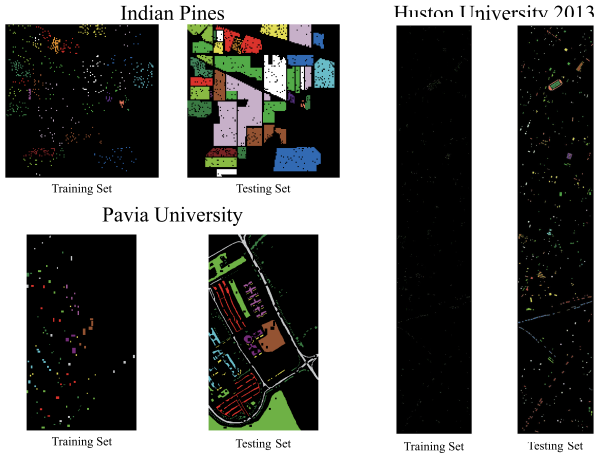


Fig. 7. Distribution of training and testing samples on the three datasets.

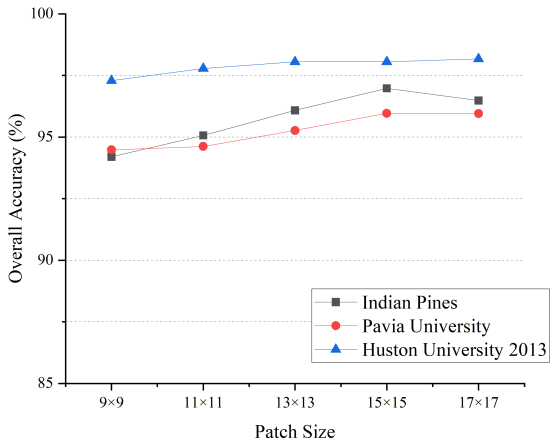


Fig. 8. Influence of patch size on proposed MRCAGCFN.

TABLE IV
DETAILED STRUCTURES OF THE PROPOSED MRCAGCFN

	CNN		GCN	
STM	Conv 1×1@96			
	Conv 1×1@48			
Feature Extraction	MRCM	MRSC@48	AFGCM	AFGC@48
		BN		BN
		Mish		Mish
	Average			
	MRCM	MRSC@48	AFGCM	AFGC@48
		BN		BN
		Mish		Mish
	Average			
	MRCM	MRSC@48	AFGCM	AFGC@48
		BN		BN
		Mish		Mish
	Average			
ALFPM	Depthwise Conv 15×15@48		Adaptive Neighborhood Aggregation	
	Average			
	Linear@B			
	Softmax			

experimental environment consists of an Intel Xeon CPU E5-2680 v4 and an Nvidia RTX 3090 GPU.

We use the Adam optimizer to train our network with a learning rate equal to 0.0005. The number of training epochs is set to 200, and the parameters with the lowest training loss are selected as the best parameters of the network for further testing. The batch sizes for the IP, PU, and HU datasets are set to 32, 128, and 64, respectively. The details of our proposed MRCAGCFN are summarized in Table IV. In this table, the number before @ represents the size of the convolution kernel, and the number after @ indicates the number of hidden layers. In addition, MRSC stands for the multiscale RSC layer, average denotes the computation of the mean from the outputs of the two preceding components, and B represents the class number.

C. Influence of Patch Size

To objectively assess the impact of patch size on our proposed MRCAGCFN, we conduct ten complete experiments for different patch sizes and then calculate the average overall accuracy (OA) as the network's performance metric. OA represents the ratio of correctly classified instances to the total number of instances in a dataset. It is calculated by dividing the number of correctly classified test samples by the total number of test samples. We experiment with patch sizes of 9×9, 11×11, 13×13, 15×15, and 17×17. The influence of patch size on our proposed MRCAGCFN is shown in Fig. 8. On the IP dataset, the network achieves the highest accuracy when the patch size is 15×15. However, on the PU and HU datasets, the highest accuracy is obtained when the patch size is 17×17. Taking into consideration network accuracy and computational cost, we ultimately select a patch size of 15×15 for the subsequent experiments.

D. Ablation Experiment

Our proposed MRCAGCFN consists of one STM and three main modules we propose. To demonstrate the effectiveness of each proposed module, we conducted experiments with various combinations and by isolating individual modules. Furthermore, to validate the efficacy of the proposed AFGCM and MRCM, we design a graph convolution module (GCM) by using a standard graph convolution layer to replace AFGC, a multiscale convolution module (MCM) by substituting RSC with standard convolution, and a multiscale deformable convolution module by replacing RSC with deformable convolution.

In all network configurations, we employed an STM as the initial step to mitigate redundant information in the spectra. Subsequently, we independently integrated our designed MRCM, AFGCM, or commonly used MCM, MDCM, and GCM to constitute the feature extraction component of the network. Finally, we utilized our designed ALFPM or global pooling to transform patch-level features into pixel-level features. The feature extraction components in all networks consist of three neural network layers. The overall structures of the networks closely align with the framework presented in Table IV.

For evaluating network performance, we adopted OA as the metric. Each network underwent ten comprehensive experiments with randomly initialized parameters, and the average OA across these experiments was calculated to provide an objective assessment of module performance. The results of the ablation experiments are presented in Table V, where

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TABLE V
OAs (%) OF NETWORKS WITH DIFFERENT MODULES ON THREE DATASETS

	Only GCN		Only CNN		Fusion	Network with ALFPM				
Module	AFGCM	GCM	MRCM	MCM	AFGCM MRCM	AFGCM MCM	AFGCM MDCM	GCM MRCM	GCM MCM	Proposed
AFGCM	✓				✓	✓	✓			✓
GCM		✓						✓	✓	
MRCM			✓		✓			✓		✓
MCM				✓		✓			✓	
MDCM							✓			
ALFPM						✓	✓	✓	✓	✓
IP	**95.905**	93.839	**96.044**	95.866	**96.318**	93.525	95.869	96.103	96.195	**96.978**
PU	**93.85**	92.074	**94.2**	94.04	**94.935**	94.663	94.43	94.344	94.295	**95.963**
HU	**97.042**	95.852	**97.396**	97.355	**97.68**	97.17	97.624	96.508	96.89	**98.057**

checkmarks (✓) indicate the modules incorporated in each configuration.

The results of experiments for only GCN indicate that, when used individually, AFGCM outperforms GCM on all three datasets. This demonstrates the effectiveness of our proposed AFGC. In addition, the separate experiments on MRCM and MCM show that the accuracy of MRCM is slightly higher than MCM, although their performances are closely comparable.

The results from the fusion network experiment, which combines AFGCM and MRCM, demonstrate that the performance of the fusion network is superior to using a single module on all three datasets. Comparing the fusion network without ALFPM to MRCAGCFN shows that ALFPM contributes to improving network performance.

We also conducted experiments by replacing the AFGCM and MRCM modules in MRCAGCFN to demonstrate the enhancement in network performance by our modules. The results of the experiments indicate that replacing any module leads to a decrease in network performance. While proving the enhancement of our module design for convolution and graph convolution, it simultaneously mitigates the imbalance in the features extracted by both methods, improving the effectiveness of feature fusion.

E. Comparison of Classification Performance

To compare the effectiveness of our proposed MRCAGCFN with other methods, we conduct experiments by using several validated methods as benchmark networks. At first, we select two traditional methods, SVM and RF. Bi-LSTM [32], which is an improved version of LSTM with attention mechanisms as a comparative method based on RNN, is also selected as a comparison method. For CNNs, we select HybirdSN [40] that uses both 2-D and 3-D convolutions to extract spatial–spectral features, A2S2K-ResNet [42] that incorporates attention mechanisms and residual networks, and GNet [44] that combines features from various layers using GNets. For GCNs, we select mini-GCN [38] that employs a subgraph training method, and SPGCN [39], which proposed a pixel-level graph convolution algorithm based on patches are selected as comparative methods. To ensure the fairness of comparative experiments, we follow the procedures outlined in the original papers and conduct multiple experiments to fine-tune hyperparameters to achieve the best results on our experimental platform. We also make efforts to utilize the source code provided by the authors of these methods. The specific configurations for some comparison models that do not have an original paper or have multiple models proposed in the original paper are as follows.

1) For the SVM, using the support vector classifier from the scikit-learn package, with a Gaussian kernel and a C value set to 100.
2) For the RF, employing the RF Classifier from the scikit-learn package, with all parameters set to default values.
3) For the miniGCN, using the fusion network with concatenation fusion strategy (FuNet-C) [38] as proposed in the article. It combines the features extracted by miniGCN with those from CNN for classification. In the original paper's experiments, it performed better than using miniGCN alone.
4) For the SPGCN, conducting experiments with SPGCN-21 [39], which demonstrated better performance in the original paper's experiments.

To quantify classification performance, we use OA, average accuracy (AA), kappa coefficient (κ), and accuracy for each class as evaluation metrics in our experiments.

AA calculates the accuracy for each class separately and then averages these accuracies. κ is a statistic that measures the agreement between observed and expected classifications while taking into account the possibility of agreement occurring by chance. It is particularly useful when dealing with imbalanced datasets. The formula for κ is as follows:

$$\kappa = \frac{p_0 - p_e}{1 - p_e} \tag{14}$$

where p_0 is the observed agreement and p_e is the expected agreement.

To ensure fairness, all performance evaluation metrics are based only on the predictions for the testing set, and we conduct ten complete experiments for all networks to calculate the average of the evaluation metrics over these 10 runs as the performance evaluation for each network.

The detailed results of the experiments on the three datasets are shown in Tables VI–VIII. The networks' classification maps for the three datasets can be found in Figs. 9–11.

From the experimental results, we can see that SVM and RF have relatively low classification accuracy on all three datasets. This is because they only use spectral information, and their

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TABLE VI
CLASSIFICATION RESULTS ON THE IP

Class No.	SVM	RF	Bi-LSTM [32]	HybirdSN [40]	A2S2K [42]	GNet [44]	FuNet-C [38]	SPGCN-21 [39]	**Proposed**
1	64.38	55.891	28.411	75.866	87.976	91.235	67.11	83.41	**92.587**
2	74.49	57.832	56.684	91.288	99.169	98.877	83.101	97.932	**99.819**
3	75	78.966	74.022	97.447	**100**	**100**	94.566	93.478	**100**
4	92.17	85.3	66.286	93.356	96.959	97.673	94.721	**98.524**	98.121
5	84.65	79.626	66.543	97.875	97.045	99.412	94.332	99.886	**100**
6	92.71	94.648	73.826	99.088	99.885	99.908	99.498	**100**	99.977
7	67.43	77.179	42.026	85.98	88.976	90.48	83.388	**95.426**	94.89
8	59.88	58.606	37.315	79.845	87.299	91.41	59.624	95.122	**96.178**
9	76.6	61.827	45.994	84.131	91.738	91.56	74.096	**94.928**	94.218
10	97.53	95.371	92.099	99.753	**100**	**100**	99.938	**100**	**100**
11	84.73	88.892	91.325	96.173	98.038	99.592	94.011	**99.696**	99.368
12	62.12	53.939	72.394	94.728	93.061	99.728	94.03	99.88	**99.97**
13	**100**	98.002	96.222	**100**	**100**	**100**	**100**	**100**	**100**
14	87.18	53.846	50.255	97.949	99.744	**100**	93.333	97.952	**100**
15	90.91	81.82	53.637	**100**	**100**	**100**	**100**	**100**	**100**
16	**100**	98	62	**100**	**100**	**100**	**100**	**100**	**100**
AA	81.86	76.234	63.066	93.344	96.243	97.492	89.484	97.266	**98.445**
OA	72.71	69.443	53.786	87.318	92.703	94.894	78.96	95.256	**96.978**
$\kappa \times 100$	69.16	65.509	49.031	85.577	91.604	94.16	76.232	94.554	**96.532**

TABLE VII
CLASSIFICATION RESULTS ON THE PU

Class No.	SVM	RF	Bi-LSTM [32]	HybirdSN [40]	A2S2K [42]	GNet [44]	FuNet-C [38]	SPGCN-21 [39]	**Proposed**
1	85.31	79.792	82.196	83.897	92.603	89.924	86.004	92.583	**93.107**
2	64.49	54.443	89.933	88.282	73.335	95.728	92.427	91.407	**96.878**
3	70.25	45.632	53.79	59.951	77.459	80.792	69.129	**86.902**	75.302
4	96.19	**98.69**	96.389	90.992	90.231	96.294	98.599	97.721	97.599
5	99.55	98.839	99.1	97.322	99.541	99.046	99.064	**99.82**	99.712
6	92.72	77.14	80.33	96.727	**99.329**	91.968	83.793	96.407	99.075
7	89.7	79.063	79.134	85.934	98.216	95.738	88.775	**99.725**	98.879
8	91.44	90.136	97.601	92.956	98.995	98.378	97.747	**99.862**	99.146
9	95.97	97.559	94.302	83.987	98.39	97.988	96.567	98.389	**98.667**
AA	87.29	80.143	85.864	86.672	92.012	93.984	90.235	**95.87**	95.374
OA	78.05	69.55	87.169	87.865	84.625	94.108	90.445	93.709	**95.963**
$\kappa \times 100$	72.41	62.277	82.912	84.231	80.487	92.09	87.227	91.653	**94.582**

TABLE VIII
CLASSIFICATION RESULTS ON THE HU

Class No.	SVM	RF	Bi-LSTM [32]	HybirdSN [40]	A2S2K [42]	GNet [44]	FuNet-C [38]	SPGCN-21 [39]	**Proposed**
1	96.46	96.01	95.96	96.096	98.988	94.673	**99.032**	96.92	99.023
2	98.57	97.247	96.758	97.548	99.834	99.622	99.446	98.255	**99.984**
3	99.55	94.894	97.657	98.958	**100**	99.85	99.97	**100**	**100**
4	97.71	94.335	95.325	90.677	**99.324**	98.739	95.598	96.85	98.468
5	98.9	98.515	96.989	99.924	99.966	99.992	99.662	**100**	**100**
6	96.43	87.598	90.227	91.527	98.118	**99.384**	87.499	99.319	95.259
7	90.61	83.612	86.013	89.9	98.124	96.952	85.849	95.082	**98.355**
8	81.29	76.724	76.291	90.55	91.212	**95.562**	87.866	92.228	95.488
9	83.52	78.25	78.537	84.77	90.814	**92.337**	80.052	87.148	91.834
10	89.01	77.614	78.858	97.76	94.164	98.584	92.008	98.37	**99.991**
11	91.9	80.155	87.809	96.896	97.496	97.323	93.692	97.604	**98.329**
12	80.87	69.871	81.409	94.493	92.058	94.261	94.784	**97.583**	97.208
13	31.01	20.044	59.864	80.157	97.796	94.968	89.033	86.246	**97.866**
14	98.28	97.462	88.817	**100**	99.902	99.926	99.038	**100**	**100**
15	98.41	93.716	95.503	99.089	**100**	99.936	99.456	99.936	99.936
AA	88.83	83.069	87.069	93.891	97.186	97.473	93.532	96.37	**98.116**
OA	90.09	84.447	87.486	94.008	96.746	97.167	93.356	96.234	**98.057**
$\kappa \times 100$	89.28	83.159	86.468	93.523	96.48	96.94	92.817	95.928	**97.902**

feature extraction capabilities are limited. On the IP and HU datasets, the performance of Bi-LSTM is even worse than traditional methods. This is likely because the high-dimensional spectral data in IP and HU datasets can lead to overfitting when treated as ordered sequential data by Bi-LSTM. Among the convolution methods, GNet performs the best, because of its novel spectral–spatial feature fusion method, which improves the balanced representation of features. FuNet-C, on the other

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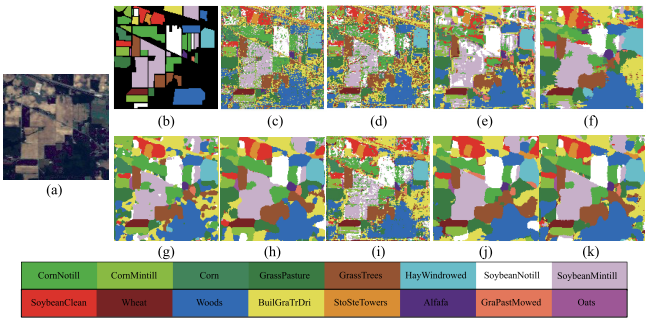


Fig. 9. Classification maps on the IP. (a) False-color image. (b) Ground-truth map. (c) SVM. (d) RF. (e) Bi-LSTM. (f) HybirdSN. (g) A2S2K-ResNet. (h) GNet. (i) FuNet-C. (j) SPGCN-21. (k) Proposed MRCAGCFN.

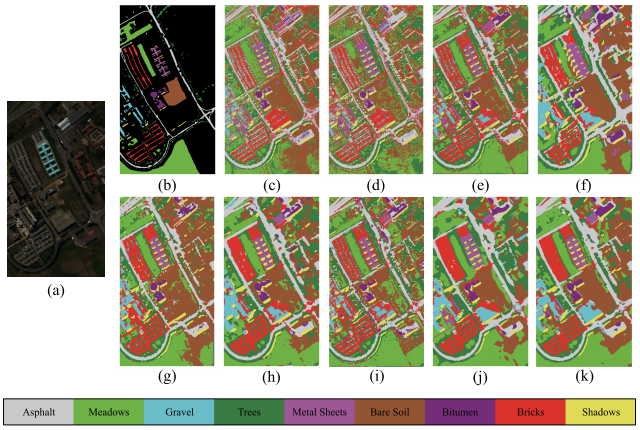


Fig. 10. Classification maps on the PU. (a) False-color image. (b) Ground-truth map. (c) SVM. (d) RF. (e) Bi-LSTM. (f) HybirdSN. (g) A2S2K-ResNet. (h) GNet. (i) FuNet-C. (j) SPGCN-21. (k) Proposed MRCAGCFN.

hand, has the poorest performance in the GCN category. This is because it only selects a small number of pixels as neighboring nodes, which results in insufficient utilization of graph structural information.

Because we employed CNN for feature fusion with patch-based GCN, our accuracy on all three datasets surpasses SPGCN. Utilizing CNN, our network exhibits stronger learning capabilities for regular-shaped features, while simultaneously alleviating the graph structure errors caused by spectral noise. The adoption of random shape convolution enhances CNN's adaptability to diverse shape features, mitigating the imbalance between its learned features and those acquired through graph convolution, thereby improving the effectiveness of feature fusion. This results in our accuracy being higher than FuNet-C, another CNN-GCN fusion network. In addition, the adaptive neighborhood aggregation method used in the GCN part enhances the neighborhood aggregation ability of the GCN, improving its spatial feature extraction capability. The ablation experiment results in Table V show that the accuracy of using AFGCM alone is higher than SPGCN. Finally, our model effectively transforms patch-level features into pixel-level features using ALFPM, further enhancing the network's accuracy.

TABLE IX
SPACE COMPLEXITY DATA OF DIFFERENT METHODS

	IP	PU	HU
Bi-LSTM [32]	1.58M	0.52M	0.88M
HybirdSN [40]	5.49M	5.49M	5.49M
A2S2K [42]	0.37M	0.22M	0.28M
GNet [44]	0.08M	0.08M	0.08M
FuNet-C [38]	0.15M	0.11M	0.12M
SPGCN [39]	**0.02M**	**0.02M**	**0.02M**
Proposed	0.61M	0.61M	0.61M

F. Analysis of the Algorithmic Complexity

In this component, we conduct an analysis of the algorithmic complexity of different methods. For space complexity, we employ the trainable parameter count as the metric, and the space complexity data of each method are presented in Table IX. As for time complexity, we utilize floating-point operations (Flops), training time, and testing time as the metrics, and the time complexity data of each method are summarized in Table X.

The experimental results reveal that SPGCN has the smallest number of trainable parameters, and FuNet-C boasts the fewest Flops, thanks to the efficiency of GCNs. This efficiency translates into a shorter training time compared to other methods. GNet also benefits from a design with small convolution kernels and PCA operation, resulting in a relatively small number of trainable parameters. Despite having fewer floating-point operations, Bi-LSTM's training time surpasses that of GNet and HybridSN due to the necessity for more iterations to achieve model convergence. Our proposed MRCAGCFN, with its adaptive feature-fusion design, larger patch size, and the fusion of convolution and graph convolution methods, entails more parameters. Nevertheless, it still maintains a lower parameter count compared to Bi-LSTM and HybridSN. The complexity of MRCAGCFN results in a substantial training time. However, it compensates for this with superior performance.

G. Influence of Different Numbers of Training Samples

To demonstrate the influence of different numbers of training samples on the experimental accuracy, we train the networks on all three datasets by using 20%, 40%, 60%, 80%, and 100% of the training samples that we have divided before. The experimental results are shown in Fig. 12.

For most networks, the decrease in the number of training samples for the PU dataset does not significantly impact accuracy. In fact, some networks achieve higher accuracy with fewer training samples. This phenomenon might be attributed to that the PU dataset is officially split, where the training set pixels are clustered together, leaving some regions in the dataset without training set pixels. As a result, the spatial information around each training set pixel and the spectral information of the training set pixels are highly similar. Therefore, a decrease in the training set ratio does not cause significant changes in the feature distribution. Moreover, since the training set itself is highly similar, converting some of it

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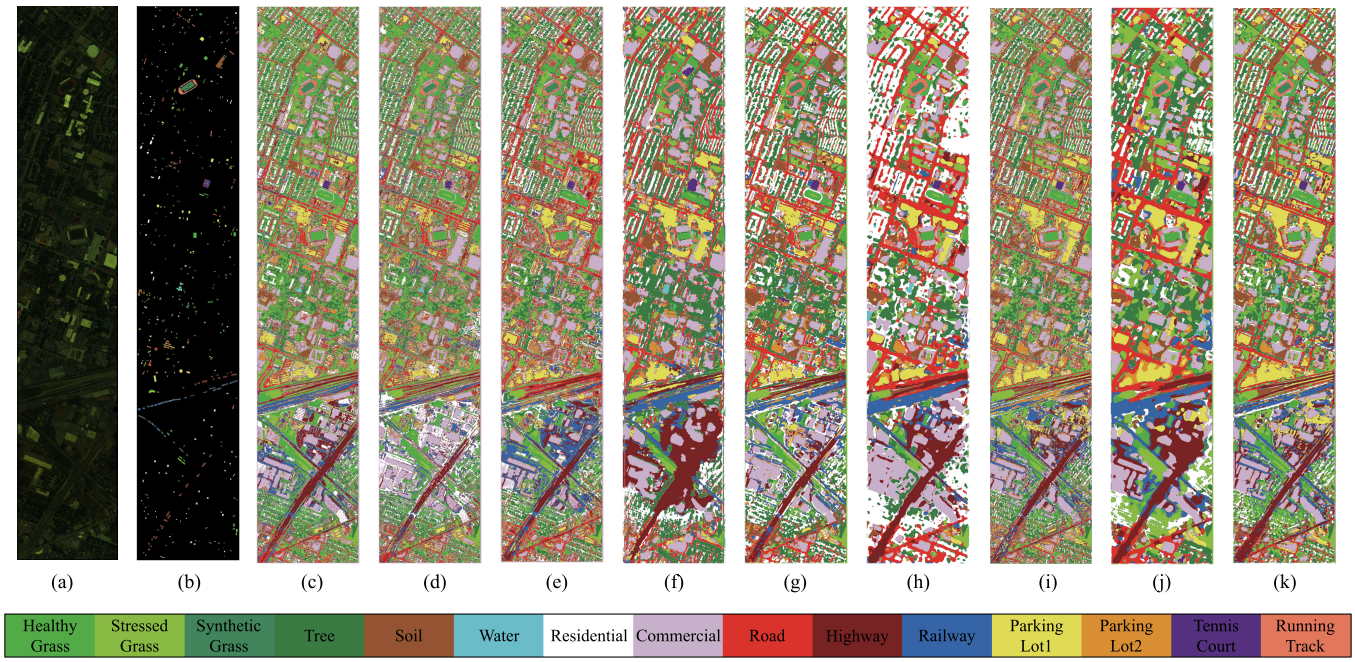


Fig. 11. Classification maps on the Houston University 2013. (a) False-color image. (b) Ground-truth map. (c) SVM. (d) RF. (e) Bi-LSTM. (f) HybirdSN. (g) A2S2K-ResNet. (h) GNet. (i) FuNet-C. (j) SPGCN-21. (k) Proposed MRCAGCFN.

TABLE X
TIME COMPLEXITY DATA OF DIFFERENT METHODS

	Method	Bi-LSTM [32]	HybirdSN [40]	A2S2K [42]	GNet [44]	FuNet-C [38]	SPGCN-21 [39]	Proposed
IP	Flops	3.08G	130.6G	5.49G	3.23G	**0.12G**	0.22G	4.31G
	Training Time(s)	265.94	110.78	366.56	142.52	99.46	**70.645**	362.04
	Testing Time(s)	**5.92**	6.57	6.97	6.78	6.33	17.54	18.6
	OA(%)	53.786	87.318	92.703	94.894	78.96	95.256	**96.978**
PU	Flops	1.25G	130.6G	2.81G	3.23G	**0.11G**	0.14G	16.97G
	Training Time(s)	1068.79	607.06	1311.7	647.01	403.3	**147.96**	1289.52
	Testing Time(s)	**15.01**	18.98	17.55	18.57	19.08	69.09	58.12
	OA(%)	87.189	87.865	84.625	94.108	90.445	93.709	**95.963**
HU	Flops	1.95G	130.6G	3.93G	3.23G	**0.09G**	0.17G	8.54G
	Training Time(s)	260.69	102.19	273.81	148.76	118.65	**71.57**	291.95
	Testing Time(s)	**7.86**	8.12	9.38	9.26	9.53	29.19	24.31
	OA(%)	87.486	94.008	96.746	97.167	93.356	96.234	**98.057**

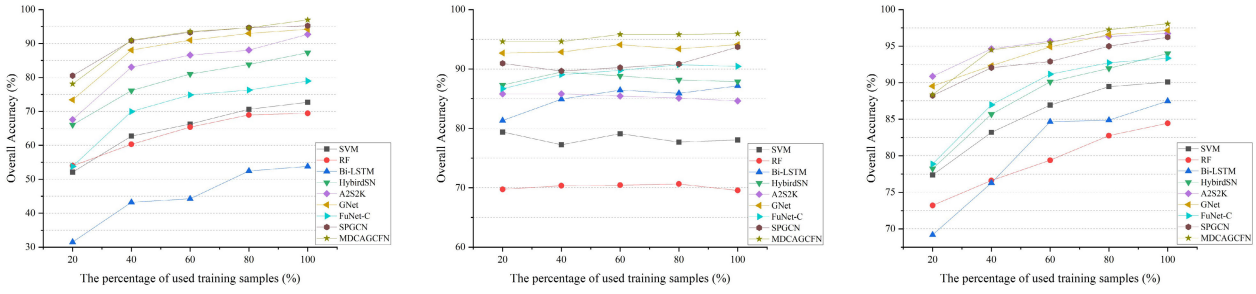


Fig. 12. Influence of different numbers of training samples on the three datasets. (a) IP. (b) PU. (c) HU.

to the testing set essentially adds easily classifiable pixels, resulting in a slight improvement in test accuracy.

In the case of the IP dataset and the HU dataset, reducing the number of training samples leads to a decrease in test accuracy. This is because the training set distribution of these two datasets is more dispersed, providing a better reflection of the overall sample distribution. When the training set ratio decreases, it causes a change in the distribution of training set features, leading to a significant drop in test accuracy.

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V. Conclusion

The shared weight property of convolution kernels endows them with strong feature extraction capabilities. However, due to the fixed shape of convolution kernels, CNNs exhibit limited adaptability to features with various shapes, which hinders their ability to address the irregularities in soil cover within HSIs.

Given the irregular nature of soil cover in HSIs, GCNs benefit from its adaptability to the underlying structures in HSI. However, the global training problem causes high costs and the need for retraining when dealing with new samples. To address these issues, patch-based GCN is applied to HSI classification, it also avoids the pixel-level feature loss caused by superpixel methods and the graph structure errors induced by mini-batch methods. However, it fails to leverage the spatial feature learning capabilities of CNN and does not effectively mitigate potential feature loss during the transformation of patch-level features into pixel-level features.

To overcome these challenges, this article presents MRCAGCFN, a patch-based CNN-GCN fusion network for HSI classification. GCNs demonstrate robust adaptability to irregular features, while CNNs excel in capturing regular spatial features. In addition, the utilization of a CNN-GCN fusion network concurrently addresses the problem of GCNs introducing incorrect features due to graph structure errors. This approach effectively harnesses the complementary strengths of CNNs and GCNs.

In the CNN component, we introduced MRCM based on RSC, enhancing CNN's adaptability to features with various shapes. Experimental results demonstrate its capability to improve the performance of our proposed network. In the GCN component, we introduced a novel AFGCM based on AFGC, surpassing traditional graph convolution methods. Finally, we employed ALFPM to address the challenge of transforming patch features into central pixel features in patch-based methods. The proposed network achieves higher classification accuracy compared to the other models we have compared against.

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